Waiting for the first success

Upper high school · probability and sequences · Synthesis problem · 08

Problem · Investigate, justify, conclude. Roll a fair die until the first 6, but stop after 10 rolls in any case. Let N be the number of rolls actually made. Find $P(N = 10)$, the probability of obtaining at least one 6, and E(N). For the mean, use $E(N) = \sum P(N > n)$, from $n = 0$ to 9.